

KEERTHIVASAN M

Cybersecurity Engineer — SOC & VAPT

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PROFESSIONAL SUMMARY

Proactive Cybersecurity Engineer specializing in **SOC operations, vulnerability management, and zero-trust architecture**. Equipped to directly enhance your organization's defensive posture by streamlining **SIEM telemetry** and deploying automated forensic workflows. My objective is to seamlessly integrate with your security operations to **reduce alert fatigue**, drive down **Mean Time to Respond (MTTR)**, and build **scalable, compliant detection frameworks** that protect critical enterprise assets and enable secure business growth.

TECHNICAL SKILLS

Security Operations (SOC): SIEM (Splunk, Microsoft Sentinel, Wazuh, ELK), Log Analysis, Incident Response, Threat Hunting

Offensive Security (VAPT): Web Application Penetration Testing, OWASP Top 10, Vulnerability Scanning

Security & Network Tools: Burp Suite, Nmap, Wireshark, Zeek, Suricata, Rapid7, Cuckoo Sandbox, SonarQube, JFrog, OWASP ZAP

Infrastructure & Cloud: AWS Security (IAM, VPC, CloudWatch), Linux Hardening, Docker, eBPF

Scripting, Automation & Dev: Python, Bash, Git, GitHub (Log parsers, API integration, custom tooling)

PROFESSIONAL EXPERIENCE

Tamil Nadu Cyber Crime Wing (TNCCW)
Cyber Security Intern (Malware Analysis & Forensics)

Chennai, India
Jun 2026 – Jul 2026

- Conducted static and dynamic analysis on malicious mobile applications and financial fraud malware to **accelerate active law enforcement investigations**.
- Automated the extraction of Command and Control (C2) domains and hardcoded IPs using custom Python scripts, **reducing manual reverse-engineering time by 80%**.
- Drafted comprehensive threat intelligence reports detailing extortion network behaviors, providing **actionable digital evidence** for non-technical investigating officers.

Zybeak Technologies
SOC Analyst Intern

Chennai, India
Jan 2025 – Feb 2025

- Monitored and triaged security alerts across cloud, network, and endpoint environments utilizing enterprise SIEM platforms including **Splunk and Microsoft Sentinel**.
- Analyzed suspicious network traffic and authentication logs using advanced network analysis tools

(Zeek, Suricata, Wireshark), escalating verified threats and recommending rapid containment strategies.

- Tuned SIEM correlation rules to filter benign anomalies, **reducing false-positive alert volume by 30%** and significantly improving overall team incident response times.

PROJECTS

NexisCore: Zero-Trust Runtime Sandbox

[\[View Repository\]](#)

- Developed a zero-trust runtime execution environment in Go, utilizing Docker and **live eBPF kernel tracepoints** to monitor and completely isolate untrusted workloads.
- Configured automated HTTP 401 unauthorization rules and SIGKILL kernel signals, successfully **blocking 100% of unauthorized network egress attempts** during simulated breaches.

Automated Malware Analysis Platform

[\[View Repository\]](#)

- Built a full-stack forensic triage application integrating Cuckoo Sandbox to facilitate **automated static and dynamic binary inspection**.
- Engineered backend data pipelines to parse memory dumps and PCAPs, generating actionable threat intelligence indicators in **under 5 minutes per sample**.

Post-Quantum TLS Readiness Simulator

[\[View Repository\]](#)

- Simulated TLS handshakes to proactively detect protocol downgrade vulnerabilities during enterprise transitions to **post-quantum cryptography**.
- Mapped cipher negotiation behaviors directly to **NIST SP 800-53 controls**, providing automated compliance scoring for legacy infrastructure.

RESEARCH PUBLICATION

Zero-Trace Biometrics & Cryptographic Enclaves

2026

Research Topic: Provably Secure Alignment-Free Minutiae Triplet Galois Fuzzy Vaults

- Formalized an end-to-end zero-trust biometric credentialing engine, eliminating single-point private key exposure by binding authentication directly to a **distributed 2-of-2 FROST threshold protocol**.
- Formulated a completely alignment-free 9-dimensional geometric minutiae representation to ensure stability against elastic physical distortions.
- Benchmarked the cryptographic pipeline on an ARM Cortex-M33 hardware enclave, successfully achieving an **Equal Error Rate (EER) of 1.15%** alongside a **sub-second execution latency of 113.13 ms**.

CERTIFICATIONS

- Google Professional Cybersecurity Certificate – [\[Verify Credential\]](#)
- AWS Cloud Solutions Architect – [\[Verify Credential\]](#)

EDUCATION

Saveetha Engineering College

Chennai, India

Student, Bachelor of Engineering in Computer Science and Engineering (Cybersecurity)

Relevant Coursework: Network Security, Digital Forensics, Cryptography, Vulnerability Assessment, OS Internals